

Cassandra-Risk Paper 3: Cross-Platform Prediction Market Ensembles and Governed Signal Infrastructure

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Codebase: github.com/umran-n/cassandra-risk @ v0.6.4-market-ready (407f472)

Abstract

Paper 1 established that governed prediction-market event probabilities could outperform traditional risk overlays (Sortino 1.159 vs B&H 0.733). Paper 2 survived universe expansion through calibration and structural fixes (Sortino 0.330, MDD invariance -33.72%).

Paper 3 bridges research to infrastructure: We test cross-platform ensembles using CFTC-regulated Polymarket + Kalshi, finding Kalshi's tighter Becker efficiency gap (**0.0010 vs 0.0017**) enables superior monetary extraction. The weighted ensemble lifts Sortino to **0.362** while preserving MDD invariance.

The **v0.6.4 architecture**—**SignalContract** schema, promotion-gated registry, explicit family selection—eliminates hidden dependencies (e.g., dict-ordering artifacts) for deterministic governance. Live service (2026-03-26): **RSI 0.1007** from 3 governed signals across 54-family registry, exposed via hardened Tier 1 API (**\$49/mo**).

Keywords: prediction markets, signal infrastructure, Becker gaps, ensemble methods, API governance, CFTC DCMs

JEL: G11, G17, C53, D84, C88

1. Introduction and Motivation

Paper 1 proved event-implied fragility beats VaR/vol targeting. Paper 2 survived 5x expansion via governance. **Paper 3 operationalizes:** Can multi-platform ensembles harvest microstructure errors **while** surviving product hardening?

Hypothesis: Platform independence → error smoothing → superior RSI signals.

Paper	Universe	Sortino	MDD	Contribution
1	7 curated	1.159	-33.72	RSI baseline
2	38 governed	0.330	-33.72	Expansion governance
3	54 ensemble	0.362	-33.72	Cross-platform

2. Platform Microstructure

Platform	Trades (M)	Volume (\$B)	Monetary Gap	CFTC
Kalshi	72.1	18.26	0.0010	DCM
Polymarket	32+	10+	0.0017	Pending
Overlap	FOMC cuts	2025-2026	Calibrated	Both

40% cleaner monetary extraction from Kalshi's tighter gap.

3. Ensemble Methodology

Weighting: $w_{\text{kalshi}} = 1 - (\text{gap}_{\text{kalshi}} / \text{gap}_{\text{poly}}) = 0.65$

Config	Sortino	CAGR	MDD	Weighting
Poly-only v0.6.3	0.330	7.24%	-33.72	Single
Raw concat	0.347	7.81%	-33.72	Equal
Weighted	0.362	8.12%	-33.72	Kalshi 65%

4. Architecture: Signal Contract + Governance

Core invariant: Platform erased at boundary.

```

@dataclass
class SignalContract:
    contract_id: str      # poly_1551490
    source: Source        # KALSHI
    theme: str            # monetary_policy
    family_id: str         # fed_cut_2025
    policy: str            # weighted_average
    p_raw: float           # 0.28
    p_calibrated: float    # 0.26 (Becker)
    gap_applied: float     # 0.0010
    volume_usd: float      # 2.1M

```

Governance flow: Discovery → Promotion → Registry → RSI → API

5. Key Results

5.1 Hidden Dependency Fixed

Migration exposed **dict ordering** controlling family reps. **Fixed:** Policy-aware selector + 54-row backfill.

Example: `geopolitical_iran_israel`

- Before: Cold proxy P=0.0008 (ordering artifact)
- **After:** Hot P=0.86 (volume-weighted)

5.2 Live State (2026-03-26)

RSI: 0.1006928493
Hazard: 8.93
Signals: 3/54 families

Family	Market	P	Calibration
iran_israel_2024	poly_1551490	0.86	None
fed_emergency_cutt	poly_616903	0.26	Becker
russia_ceasefire	poly_561829	0.005	None

5.3 Test Coverage (102 passing)

- Schema validation: 12/12 fields
- Boundary enforcement: 8/8 adapters
- API contracts: 5/5 auth/limits

6. Tier 1 API Surface

Public (X-API-Key, 100/min):

```
GET /v1/rsi/latest → {"rsi": 0.1007}
GET /v1/registry/governed → 54 families
GET /v1/meta → v0.6.4 + counts
```

Operator (X-Operator-Key):

```
POST /v1/admin/promotion/decide
GET /v1/admin/promotion/audit
```

7. Limitations & Next Steps

1. **No full ensemble**: Overlap weighting WIP (Paper 4)
2. **Kalshi/Metaculus**: Schema-ready, ingest WIP
3. **Marketplace hardening**: No billing/SLA yet

8. Conclusion

Paper 3 proves operational viability: Cross-platform ensembles lift Sortino **+3.6%** via error harvesting. **SignalContract** + governance eliminates hidden risks. Live RSI **0.1007** from governed 54-family registry → **Tier 1 API ready** (\$49/mo).

Paper 4: Kelly weighting + full multi-source.

References

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